



Infineon/BIT Excellence Program 2026
Sustainable Supply Chain for Dried Mango from Africa to Germany

Module 05
Energy Supply Systems Management

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1. Energy Supply Systems: Scope Definition and Component Registry

1.1 Scope and Perimeter of Work

The **Energy Supply Systems module** is built to handle everything related to power on the farm generation, storage, distribution, and monitoring. It brings together all sources of electricity, from solar panels to the backup diesel generator, into one place. Instead of assuming energy is just a fixed input for each production module, this system creates a central registry of every power asset and flow. That way, the farm can track production, consumption, costs, and carbon impact separately from the processes that use the energy.

In scope are all the assets and data streams directly tied to producing, storing, and distributing electricity: the solar PV array, battery bank, inverter, diesel generator, and the monitoring/IoT tools that keep track of them. For each of these, the module focuses on six areas: monitoring solar generation, tracking energy use across farm operations, managing renewable energy sizing, optimizing costs, calculating carbon footprint, and managing generator backup.

What's **not included** are the physical machines that consume the energy. For example, the irrigation pump itself and the decision of when to run it belongs to the Water Supply module. Here, we only log the pump's power draw and operating hours. The dryer, grinder, and other processing equipment are part of the Product Transformation and Machinery modules; this system only records their energy demand, not their maintenance or ownership. Likewise, tractors, motor pumps, and other mechanized equipment are tracked under the Machinery module, while vehicles transporting dried mango to Germany fall under Sales and Marketing.

The key distinction is between **owning a load** and **owning the power that feeds it**. The Water Supply module manages irrigation schedules and water quality. The Product Transformation module ensures drying quality. The Machinery module owns the physical equipment. The Energy Supply Systems module, by contrast, is responsible only for how much power each load consumes, where that power comes from (solar, battery, or generator), and its cost both financial and environmental. This separation avoids duplicating scheduling or maintenance data while still giving full visibility into every watt the farm uses.

One boundary still under discussion with the Machinery module owner is how to log diesel fuel for the generator. Should it appear only here, under the Generator Events table, or also in Machinery's Fuel Log (which currently covers tractors and motor pumps)? For now, the assumption is that generator fuel is tracked exclusively in this module, since the generator is considered an energy-generation asset rather than farm machinery.

1.2 Data Consolidation from 2025 Cohort

Before setting up this year's component list and load model, we reviewed the 2025 cohort's Excel file specifically Abdoul's Solar PV Power Plant sheet as a baseline. This followed the instruction to build on last year's sustainable farm design rather than starting from scratch.

In that 2025 baseline, the irrigation pump was defined at 1.5 kW with an assumed 4-hour daily runtime, plus miscellaneous loads of 3 kW over 5 hours. Once this year's Water Supply

module published real irrigation figures (20 m³/day at a flow rate of 4 m³/h), the pump's operating time was corrected to 5 hours. That adjustment became one of the first resolved inconsistencies between last year's assumptions and this year's validated data.

An early draft of this module (see Tables 1–8 in the working file) sized the system around a single fixed scenario of 37 kWh/day. That breakdown included irrigation (17), ventilation/greenhouses (7), storage/cooling (5), tools/machinery (4), electric fences (2), and office/management (2). However, this fixed-scenario model produced a payback period of 31–32 years too long to be a credible business case.

It was replaced by a scenario-based model aligned with the Product Transformation module's production tiers: Base, 50 kg/day, 200 kg/day, and 1,000 kg/day of dried mango. This way, energy sizing scales with the same production volumes used elsewhere in the farm's business case. Four new load categories were added beyond the 2025 baseline to reflect farm-wide needs not previously covered: storage/cooling (8 kWh/day), lighting (2 kWh/day), security (fencing and cameras, 1 kWh/day), and monitoring/IoT (0.5 kWh/day). Together, these add 11.5 kWh/day. Combined with the corrected pump load (7.5 kWh/day) and Abdoul's miscellaneous load (15 kWh/day), the Base scenario now totals 34 kWh/day before any mango drying activity. At the 1,000 kg/day tier, consumption rises to 3,034 kWh/day once dryer demand is included.

The guiding principle for this year's Energy Supply Systems module is clear: every generation and storage asset is tracked once, under a single component registry, while consumption is modeled per load category and scaled against the farm's actual production scenarios. This avoids the pitfalls of fixed assumptions and ensures the energy model grows in step with the farm's operations.

1.3 Consolidated Component List

The table below lists the physical energy assets tracked by this module, with fictional component IDs assigned for use in this year's prototype, following the same MCH-XXX convention used by the Machinery module (Module 08), adapted here as ENG-XXX.

Component ID	Component	Category	Source
ENG-001	Solar PV array (polycrystalline)	Generation	Abdoul (2025), revised Aida
ENG-002	Battery storage bank (Li-ion)	Storage	Aida (2026)
ENG-003	Inverter	Power conversion	Aida (2026)
ENG-004	Backup diesel generator (8 kW)	Backup generation	Aida (2026)
ENG-005	Monitoring / IoT controller	Monitoring	Aida (2026)

Component IDs follow the pattern ENG-XXX and are entirely fictional at this stage, meant to populate the frontend prototype rather than reflect real serial numbers. Once the farm is operational, each ID would be replaced by a physical asset tag linked to the component's actual purchase and installation record.

1.4 Data Model

The data model below turns the six focus areas of this module generation monitoring, consumption tracking, sizing and cost, carbon accounting, generator backup, and alerting into five linked tables, all built around the Energy Component table as the central reference. Every other table refers back to a component through its `component_id`, which keeps the model consistent and avoids repeating asset details in each log. This structure is what will be used to generate the fictional dataset behind the frontend prototype.

Energy Component

Field	Type	Description
<code>component_id</code>	string (PK)	Unique identifier, e.g. ENG-001
<code>name</code>	string	Component name
<code>category</code>	enum	PV Array / Battery / Inverter / Generator / Monitoring
<code>capacity_value</code>	decimal	Rated capacity (kWp for PV, kWh for battery, kW for inverter/generator)
<code>purchase_date</code>	date	Date the component entered service
<code>unit_cost_eur</code>	decimal	Purchase cost in euros
<code>status</code>	enum	Operational / Under maintenance / Out of service
<code>location</code>	string	Site where the component is currently based

Solar Generation Log

Field	Type	Description
<code>log_id</code>	string (PK)	Unique identifier for the generation entry
<code>component_id</code>	string (FK)	Reference to the PV array in Energy Component
<code>date</code>	date	Date of the reading
<code>kwh_produced</code>	decimal	Energy produced that day
<code>efficiency_pct</code>	decimal	Panel output relative to rated capacity

Consumption Log

Field	Type	Description
<code>log_id</code>	string (PK)	Unique identifier for the consumption entry
<code>date</code>	date	Date of the reading
<code>load_category</code>	enum	Irrigation / Storage-Cooling / Lighting / Security / Monitoring / Processing
<code>kwh_consumed</code>	decimal	Energy consumed that day by this load category
<code>source</code>	enum	Solar / Battery / Generator / Grid

Generator Events

Field	Type	Description
event_id	string (PK)	Unique identifier for the backup event
date	date	Date the generator was activated
trigger_reason	text	Why the generator started, e.g. battery below threshold
duration_hours	decimal	Hours the generator ran
fuel_liters	decimal	Diesel consumed during the event
cost_eur	decimal	Cost of that event, fuel included

Carbon Metrics

Field	Type	Description
metric_id	string (PK)	Unique identifier for the carbon entry
period	date	Month or reporting period covered
co2_avoided_kg	decimal	CO2 avoided versus grid baseline for that period
emission_factor_used	decimal	kg CO2 per kWh applied for that source
scenario	string	Production scenario this reading relates to (Base / 50 / 200 / 1,000 kg/day)

Alerts

Field	Type	Description
alert_id	string (PK)	Unique identifier for the alert
component_id	string (FK)	Reference to the Energy Component table
severity	enum	Critical / Warning / Info
message	text	Description of the alert, e.g. battery below 20%
timestamp	datetime	When the alert was raised
status	enum	Open / Resolved